

continuities) are greater than 7.6cm (3-inch), reflected pulse will arrive at point Y. Since driven clock is now stable, reflected pulse will have significant effect on incident clock.

Hence, for a PCB with dimensions greater than 7.6cm (3-inch), clock with rise times of 1ns will be considered high-speed and, for such cases, a radical design approach termed as high-speed PCB design using multi-layer boards will have to be followed.

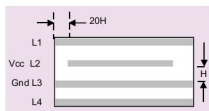


Fig. 7 (a): The 20H rule

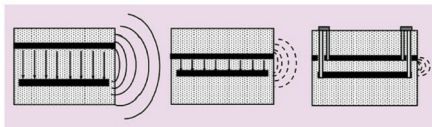


Fig. 7 (b): Reducing fringing fields

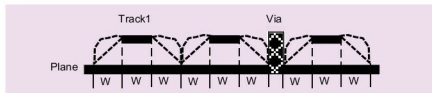


Fig. 8: The 3W rule

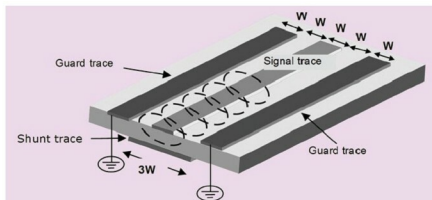


Fig. 9: Guard and shunt traces

Multi-layer concepts

Let us take a look at some basic terms related to multi-layer boards that we will be referring to time and again.

Board or layer stack-up. Stack-up is the name given to the order of various etched copper and dielectric layers that make up the board.

Power and ground planes. Multi-layer boards invariably use two layers of un-etched copper that are in the form of a continuous sheet, one for power supply (Vcc) and the other for ground (return), which are normally referred to as power and ground planes. Use of individual planes for power and ground lowers power distribution impedance and also reduces the signal-loop area. Hence, compared to a two-layer PCB, a simple four-layer PCB using such signal distribution can provide

approximately 30dB improvement in EMC performance.

Reducing fringing fields: The 20H rule

Radio frequency (RF) currents between power and ground planes will cause emissions at the edges of planes. Such emissions are sometimes referred to as edge-fired emissions or fringing fields. These can be reduced by reducing the size of the power plane with respect to ground plane.

Ground plane dimensions [on layer 3 in Fig. 7(a)] should exceed power plane dimensions by 20H on all four sides, where H is the distance between power and ground planes. This will increase intrinsic self-resonant frequency of the PCB. Decreasing spacing between 0V and power planes also reduces fringing fields. Further decrease can be achieved by using a decap wall around the periphery of the power plane and inside the 0V plane periphery by 20H, where H is the plane pair spacing.

Reducing cross-talk: The 3W rule

Cross-talk is capacitive coupling of signals from one circuit track to another. This can occur when a high-frequency track is running parallel to a susceptible track. Cross-talk flux and associated capacitive coupling can be reduced by 70 per cent if the two tracks are separated by a distance of 3W, where W is the width of the traces (measured from trace centre). This will cause track flux to terminate on the ground plane below rather than on adjacent tracks (shown as dotted lines in Fig. 8).

Guard traces

Traces carrying high-frequency signals like clocks can cause coupling of clock harmonics on nearby traces, causing a jittering clock that can lead to problems elsewhere in the circuit. These problems can be reduced by providing guard traces (Fig. 9) on either sides of such a track. Distance